1. Primary Alignment: Items 4, 7, and 5

- **Item 4: Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation**
  - **Analysis:** This is the most critical step. Students must decode the text ("sealed space," "air never flows freely") and interpret the diagrams (specifically the one showing "possible air flow") to construct a physical model of the sealing condition (p. 1). They must understand that for the door to be sealed, one wing must always reach the edge of an opening before the previous wing leaves the other edge.
- **Item 7: Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis**
  - **Analysis:** A revolving door is a dynamic, moving system. To solve this, students must "simplify" the motion into a critical state: realizing that the maximum opening is limited by the size of a single sector ($120°$). Decomposing the continuous rotation into a static geometric relationship between arc lengths is essential for analysis (p. 1).
- **Item 5: Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations**
  - **Analysis:** Students must recognize the underlying geometric structure: three wings divide the circle into three equal $120°$ sectors. The key "relationship" to discover is that the arc length of the opening cannot exceed the arc length of one sector (p. 1).

2. Significant Alignment: Items 6, 8, and 12

- **Item 6: Identify and describe the mathematical aspects of a real-world problem... identifying the significant variables**
  - **Analysis:** Students identify the significant variables: the diameter ($200\ \mathrm{cm}$), the number of wings (3), and the resulting circumference. They must describe "sealing" in terms of these mathematical variables (p. 1).
- **Item 8: Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures**
  - **Analysis:** The student relates the real-world "opening" to the known mathematical concept of **circumference** ($C = \pi d$) and **proportionality** (calculating $1/3$ of the total circumference) (p. 1).
- **Item 12: Identify the constraints, assumptions simplifications in a mathematical model**
  - **Analysis:** The problem provides a strict **constraint**: "air never flows freely." Students must work within this constraint, assuming a simplified model where the wings and the circular space fit perfectly with no gaps (p. 1).

1. 核心关联：第 4 项、第 7 项与 第 5 项

- **第 4 项：Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation (p. 1)**
  - **分析：** 这是解题的第一步，也是最关键的一步。学生必须通过阅读文字说明（“sealed space”、“air never flows freely”）并解码复杂的俯视图，来构建一个关于“密封条件”的物理模型。学生需要理解，当一扇门翼到达开口的一端时，另一扇门翼必须已经覆盖了开口的另一端。
- **第 7 项：Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis (p. 1)**
  - **分析：** 现实中的旋转门是一个动态过程。为了进行数学分析，学生必须将这个动态过程“分解”或“简化”为一个临界状态：即每个扇区（$120°$）必须始终能挡住开口。这涉及到将复杂的物理运动简化为圆周几何中的弧长关系。
- **第 5 项：Recognise mathematical structure (including regularities, relationships, and patterns) in problems or situations (p. 1)**
  - **分析：** 学生需要识别出题目背后的几何结构：三个相等的扇区（每个 $120°$ 或圆周的 $1/3$）。解决问题的关键在于发现“最大开口弧长不能超过单个扇区的弧长”这一数学规律。

2. 重要关联：第 6 项、第 8 项与 第 12 项

- **第 6 项：Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables (p. 1)**
  - **分析：** 学生需要识别出影响密封性的关键数学变量：圆的直径（200 cm）、门翼的数量（3个）以及由此产生的扇区弧长。
- **第 8 项：Recognise aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures (p. 1)**
  - **分析：** 学生需将“密封”这一现实问题转化为已知的数学概念——**圆周长公式** ($C = \pi d$) 和**比例计算**。他们必须意识到，最大开口弧长实际上就是圆周长的 $1/3$。
- **第 12 项：Identify the constraints, assumptions simplifications in a mathematical model (p. 1)**
  - **分析：** 题目本身设定了一个严格的约束条件（Constraint）：空气永不自由流动。学生必须在这一约束下进行建模，并假设门翼与圆形空间之间是完美契合的简化状态。